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# NECK MUSCLE FATIGUE AFFECTS POSTURAL CONTROL IN MAN

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**Abstract—We hypothesised that, since anomalous neck proprioceptive input can produce perturbing effects on posture, neck muscle fatigue could alter body balance control through a mechanism connected to fatigue-induced afferent inflow. Eighteen normal subjects underwent fatiguing contractions of head extensor muscles. Sway during quiet stance was recorded by a dynamometric platform, both prior to and after fatigue and recovery, with eyes open and eyes closed. After each trial, subjects were asked to rate their postural control. Fatigue was induced by having subjects stand upright and exert a force corresponding to about 35% of maximal voluntary effort against a device exerting a head-flexor torque. The first fatiguing period lasted 5 min (F1). After a 5-min recovery period (R1), a second period of fatiguing contraction (F2) and a second period of recovery (R2) followed. Surface EMG activity from dorsal neck muscles was recorded during the contractions and quiet stance trials. EMG median frequency progressively decreased and EMG amplitude progressively increased during fatiguing contractions, demonstrating that muscle fatigue occurred. After F1, subjects swayed to a larger extent compared with control conditions, recovering after R1. Similar findings were obtained after F2 and after R2. Although such behaviour was detectable under both visual conditions, the effects of fatigue reached significance only without vision. Subjective scores of postural control diminished when sway increased, but diminished more, for equal body sway, after fatigue and recovery. Contractions of the same duration, but not inducing EMG signs of fatigue, had much less influence on body sway or subjective scoring. We argue that neck muscle fatigue affects mechanisms of postural control by producing abnormal sensory input to the CNS and a lasting sense of instability. Vision is able to overcome the disturbing effects connected with neck muscle fatigue.** **© 2003 IBRO. Published by Elsevier Ltd. All rights reserved.**

**Key words: fatiguing contractions, EMG, posture, body sway, subjective score.**

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*Abbreviations:* aF1, after the first fatiguing period; aF2, after the second fatiguing period; aR1, after the first recovery period; aR2, after the second recovery period; CFP, centre of foot pressure; EC, eyes closed; EO, eyes open; F1, first fatiguing period; F2, second fatiguing period; MVC, maximal voluntary contraction; R1, first recovery period; R2, second recovery period; SA, sway area; SP, sway path.

There are many indications that the afferent input from neck muscle plays a role in the control of stance and locomotion. Studies using natural neck proprioceptive stimulation (head rotation relative to the trunk) have shown that neck afferent inflow is important in perceiving not only head posture (Mergner et al., 1991) but also motion and localisation of visual objects (Mergner et al., 1992). Other investigations used neck muscle vibration as an adequate stimulus to elicit proprioceptive input. Vibration in normal subjects produced 'compensatory' adjustments of limb and body position, illusory or actual movements of the head or whole body and illusory displacement of a visual object being viewed by the subject (Goodwin et al., 1972; Biguer et al., 1988; Roll et al., 1991; Taylor and McCloskey, 1991). When applied during stance, vibration produced increased body sway (Koskimies et al., 1997) and an inclination of the body toward the side opposite to the vibrated muscle (see Kavounoudias et al., 1999). Neck muscle vibration also modifies the representation of visual motion and direction (Biguer et al., 1988; Karnath et al., 1994, 2002). Further, during stepping in place, long-lasting lateral vibration induces body rotation toward the contralateral side (Bove et al., 2002). Not unlikely, the above effects may be linked to a cervical proprioceptive input to the brain stem nuclei which also control posture, e.g. the vestibulospinal nuclei (Boyle and Pompeiano, 1981; Mazzini and Schieppati, 1994; Gdowski and McCrea, 2000). Neck torsion has an influence on the direction of body displacement induced by galvanic vestibular stimulation (Gurfinkel et al., 1989; Fransson et al., 2000), and vestibular-proprioceptive interactions play a role in self-motion perception in addition to postural control (Mergner et al., 1997).

There are indeed suggestions that abnormal cervical inflow may cause dizziness or imbalance, sometimes termed 'cervical vertigo' (see for a review Brandt and Bronstein, 2001). Objective assessment of stability using stabilometric techniques has given indications that these pathological conditions may or may not produce increased body sway in these subjects (Alund et al., 1993; Karlberg et al., 1996). However, measures of sway were obtained during quiet stance with the head in the primary position. Patients hardly complain of dizziness under these conditions, whilst it is abnormal head posture that can trigger episodes of dizziness (Endo et al., 2000). It is therefore possible that the cause of imbalance is not the pathological process per se, but the consequences of such a process that may occur under particular circumstances. For instance, uncomfortable head position or prolonged neck muscle contraction could possibly favour the appearance of dizziness through a mechanism connected to an abnormal fatigue-related afferent inflow. Muscle fatigue can in-

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deed affect perception of body segment position (Sharpe and Miles, 1993; Carpenter et al., 1998; Taimela et al., 1999; see for a recent review Taylor et al., 2000). The hypothesis was therefore put forward that neck muscle fatigue could be a cause of poor stance control. To this end, we produced fatigue in normal subjects by intense, prolonged isometric contraction of the dorsal neck muscle, and prior to and immediately after this procedure we recorded the body sway during stance. The subjective evaluation of body sway (Schieppati et al., 1999) was also collected, in order to assess any effect of neck muscle fatigue on the perception of stance stability. We concluded that fatigue indeed produces a transient increase in body sway accompanied by a feeling of instability. A preliminary communication has been presented (Beretta et al. 2002).

## EXPERIMENTAL PROCEDURES

### Subjects

Eighteen healthy subjects (11 men and seven women, mean age 35.5 years, range 23–54) volunteered for these experiments. No subject had a history of neck pain or of illnesses known to produce imbalance. No had symptoms or signs of neck muscle or cervical column disease or had suffered head trauma or whiplash injury. Informed consent was obtained according to the Declaration of Helsinki, although the specific aims of the experiment were not conveyed. The local Ethics Committee approved the investigation and experimental procedures.

### Balance assessment

Body sway during quiet upright stance was recorded using an AMTI (model SGA6, Watertown, MA, USA) dynamometric platform. Each trial lasted 60 s; during this period subjects were asked to stand upright and barefoot on the platform as still as possible with arms by their side. Feet were parallel, the distance between the two internal malleoli being about 1 cm. Subjects were tested with eyes open (EO), when they looked at a target placed at eye level about 1.5 m in front of them, and with eyes closed (EC). Two successive trials for each condition were performed, except for the first block (control conditions), for which four trials per visual condition were acquired. The platform force signals were converted from analog to digital, at a sampling rate of 100 Hz (low-pass filtered at 1050 Hz), and fed into a computer. The following variables were computed off-line: (1) position of the instantaneous centre of foot pressure (CFP), (2) sway path (SP) or distance covered during the trial by the moving CFP, (3) sway area (SA) or the area enclosed by the path of movement calculated as the surface swept by the line joining the mean CFP to the instantaneously moving CFP. Since SP is linearly and positively correlated to SA (Schieppati et al., 1993), we analysed SA alone as a synthetic measure of body sway. Since SA values are not distributed in a normal fashion, the log10 of the SA value was considered for statistical analysis. Body SA was rather variable from trial to trial, within and across subjects. The values to be subjected to further analysis were the average values within blocks and visual condition (see below) for each subject.

### Subjective evaluation of quiet stance

In accordance with a recently developed method (Schieppati et al., 1999), at the end of each stabilometric trial patients were asked to score their own balance performance. Scores ranged from 10 ('I felt really still as if holding onto a stable frame') to 0. To further illustrate, a medium score (5) would correspond to the sensation felt by a normal subject standing upright with one foot placed in front of the other on the same line; zero could be scored by a subject standing on one leg with EC without support, and eventually putting down the other foot.

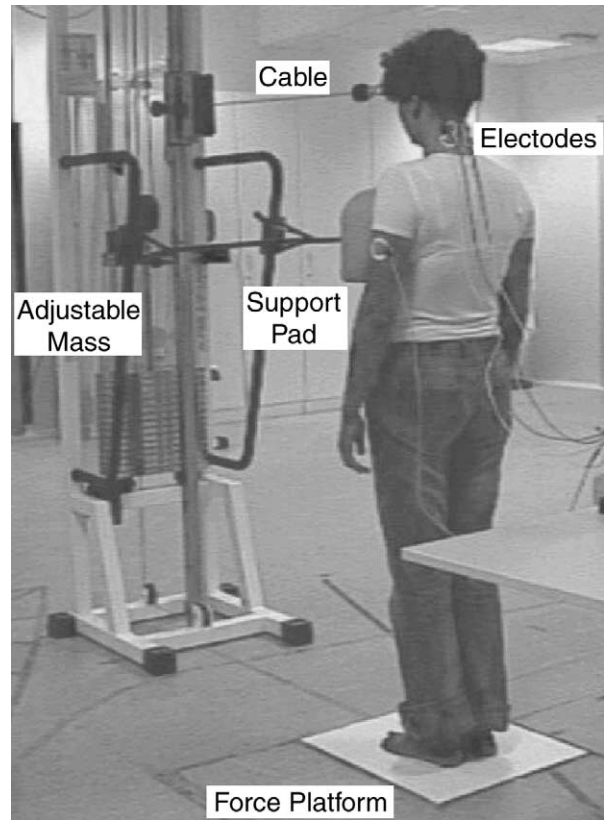


**Fig. 1.** Position of the subject during the fatiguing procedure of the dorsal cervical muscles. The subject had the same posture as during the stabilometric trials, with the feet on the force platform. The extensor torque was exerted against a weight stack attached to a cable of a standard training column. Head posture was maintained in the primary position through visual feedback of the vertical position of a marker fixed to the cable. The subject leant against a padded support placed at sternum height.

### Fatiguing procedure

Subjects were put on the stabilometric platform, in the normal upright position (Fig. 1). A large belt was passed around the head just above the ears, and a cable fixed to it frontally and medially in correspondence with the forehead. The cable passed through a pulley fixed at 1-m distance at head height. The pulley was part of a versatile gym training column placed in front of the subject. Its vertical position was adjustable so that the line of the pulling action was horizontal. The other end of the cable was fixed to an adjustable mass. The thorax of the subject leant against a padded vertical support firmly fixed to the cable column, so that the action of the masses did not displace the upright body posture. No contraction of the body extensor muscle chain below the level of the support was required, since there was no need to push the trunk or whole body backwards, owing to the thorax support, which nullified the body forward-pulling action of the mass. Therefore, the muscle action was broadly limited to the neck extensors, and the subject counteracted the head-flexor torque exerted by the mass by isometrically contracting the neck dorsal muscles. The subject had visual feedback of the head position by way of a target fixed to the cable, which had to remain co-planar with a reference fixed to the column. This was sufficient enough to

ensure negligible head pitch during neck muscle contraction against the head-flexor torque produced by the mass. For each subject, the mass was chosen so that it represented about one third of the maximal voluntary contraction (MVC, 47.6±11.0 kg), i.e. 16.4 kg±5.3 S.D., equal to 34.9±9.5% of MVC. MVC was the highest force value exerted during a 10-s period of neck extension, during which the subject was encouraged to produce the highest possible effort. This was measured by means of a portable dynamometer (Chatillon, type CSD 500, Ametek Inc., Paoli, PA, USA) placed in series between the head belt and the pulley. During the fatiguing periods, the procedure was that the subject had to counteract the applied mass for 5 min.

### Sequence of stance trials

After the acquisition of eight stance trials (four with EO and four with EC) under control conditions, subjects underwent the first fatiguing period (F1), performing the tonic, prolonged neck muscle contraction intended to produce neck muscle fatigue. After F1, four subsequent quiet-stance trials (alternating EC and EO) were recorded. Less than 10 s elapsed from when the mass was removed to the beginning of the first sway recording. During this period there was no need to correctly re-position the subject on the platform, since no perturbation to the vertical body and head posture was produced when the mass was taken away. Following this acquisition block, the connecting wires were disconnected from the EMG electrodes, and subjects were free to sit or move around the laboratory for 5 min. This period was the first recovery (R1) period, and was followed by four more stance trials. A second fatiguing period (F2) followed, in turn followed by four more stance trials. The protocol during these sequences exactly replicated F1 and after-F1 (aF1). A final period of 5-min recovery (R2) preceded the last four stance trials. The order of trials was always the same for all subjects.

### EMG recording

Electrical activity of the dorsal neck muscles was recorded in 13 subjects (from both right and left side) using bipolar pre-jelled surface electrodes, with a longitudinal distance of 1 cm between the recording spots. The electrode pairs were placed on the bellies of the dorsal neck muscles, 2 cm from the midline and 4 cm below the cranial insertion. The electrodes were left in place during the entire recording session. The signals were differentially amplified (×1000), filtered (cutoff frequency 500 Hz) and acquired on a PC (sampling frequency 1 kHz) using a Biopac (Santa Barbara, CA, USA) amplification and recording system. The EMG signals were acquired for the duration of 10 s from the beginning of each stance trial performed during the session. During the prolonged (fatiguing) isometric neck extensions (F1 and F2 periods), EMG activity was recorded during an initial 10-s period of each minute (periods lasted 5 min). The analysis of the EMG was conducted by means of the Acknowledge software, which gave the amplitude of the filtered and integrated signal (the area of the envelope) and its median power frequency over the recorded epochs (Merletti and Lo Conte, 1997; Yoshitake et al., 2001).

### Statistical analysis

To test the significance of the differences across the mean SAs or mean positions of CFP, recorded during the five quiet-stance blocks, under the two visual conditions, a two-way ANOVA (visual conditions as independent variables and stance blocks as repeated measures) was used. Analogously, the EMG variables (median frequency and amplitude) during the quiet stance trials were subjected to two-way ANOVA. The same procedure was used to assess the differences across the neck EMG variables recorded during the 5 successive minutes of the fatiguing procedure (F1 and F2 as independent variables and EMG variables recorded during each minute of muscle contraction as repeated measures). When appropriate, the Newman-Keuls test was used for post hoc comparisons. The level of significance at which the null hypothesis was rejected was equal to $P<0.05$.

## RESULTS

EMG data will be presented first, followed by sway data. For both variables, the values obtained during the standing trials are evaluated in sequence: control, aF1, after first recovery period (aR1), after second fatigue period (aF2), after second recovery period (aR2), in turn separated into visual conditions (EO; EC). In addition, the EMG data for F1 and F2 are both considered, in order to establish whether the contractions produced in these two periods showed electrical signs of fatigue.

### EMG during fatiguing contractions

The neck muscle EMG recorded during the 5-min fatiguing contractions (F1 and F2) showed a progressive diminution of the median frequency and a progressive increase in amplitude in all subjects, on both right and left side.

### Signal median frequency

On average, at the fifth minute of contraction, the median frequency progressively decreased to about 80% of its value at the first minute of the first fatiguing contraction (F1; Fig. 2A). These progressive changes were highly significant ($F=33.2$; df 4,192; $P<0.001$). The post-hoc test indicated no effect of side (left versus right) but an effect of sequence: at every successive minute the decrease in EMG median frequency was significant (post hoc test, $P<0.01$), except between the third and fourth, and the fourth and fifth minute, indicating a trend to plateauing. The findings during the F2 fatiguing period were super-imposable to and not significantly different from those observed during F1. Such EMG behaviour is evidence of the occurrence of physiological changes characteristic of muscle fatigue (Vøllestad, 1997).

### Signal amplitude

On average, at the fifth minute of contraction the EMG increased to about 130% of its value at the first minute of F1. Grand averages of the area of the envelope of integrated and filtered EMG recorded during the F1 and F2 fatiguing periods are reported in Fig. 2B. Analysis of variance indicated that the progressive increase in amplitude was significant (ANOVA, $F=17.3$; df 4,192, $P<0.001$), whilst there was no difference in the effects produced by F1 with respect to F2. The post hoc test indicated that the increase in EMG amplitude was significant, between the first and second minute and between the second and third. The progressive increase in EMG activity is the other sign of development of neck muscle fatigue during the 5-min period of isometric muscle contraction against the load.

### EMG during quiet stance

EMG during quiet stance, with the neck muscles at rest, was recorded both prior to (control) and after the fatiguing con-

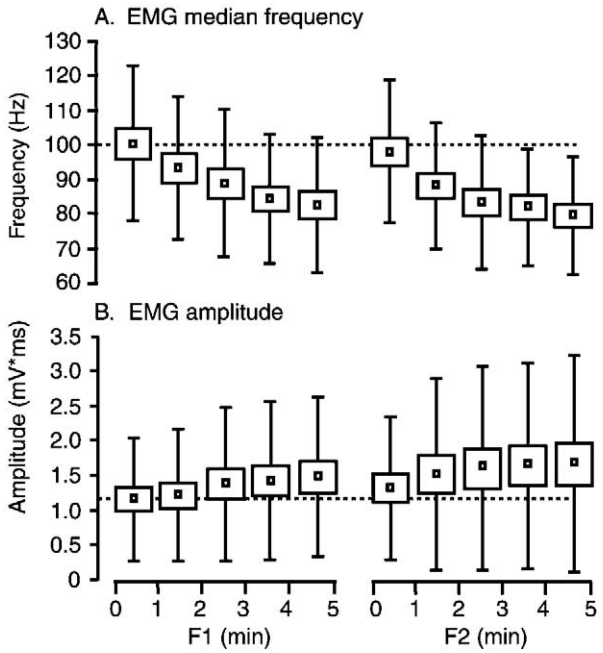


**Fig. 2.** EMG during fatiguing contractions. Median frequency and amplitude changes of the EMG of the dorsal cervical muscles (ordinate) recorded at each minute during the 5-min F1 and F2 (abscissa). A. Median frequency progressively and significantly diminished during F1. At the beginning of F2, the median frequency had recovered, to progressively diminish with a time-course similar to F1. B. Amplitude progressively and significantly increased during F1 and F2, with a time-course similar to that of the corresponding median frequency changes. Mean values (±S.E.M., boxes, and ±S.D., whiskers) from all subjects, data from right and left muscles collapsed.

tractions (aF1 and aF2) and recovery (aR1 and aR2), simultaneously with the acquisition of the force-platform stabilometric data. The EMG of the dorsal neck muscles was small in amplitude (less than one tenth of that produced when supporting the load) and symmetric between the right and left sides (*t*-test, *ns*), and unaffected by the visual condition (*t*-test, *ns*) in any subjects. The median frequency of the traces was rather variable across subjects and sides; it was somewhat greater (by about 30%) under resting conditions compared with when supporting the load. On average, amplitude and frequency of the EMG signal (side and visual condition collapsed) showed very small but systematic changes during the several stance acquisitions recorded before and after the fatiguing procedure. Median frequency and mean amplitude showed in fact moderate decreases (ANOVA, $F=11.7$; df 26,104; $P<0.001$) and increases (ANOVA, $F=10.3$; df 26,84; $P<0.001$), respectively, after both F1 and F2. EMG almost returned to control values after the recovery periods (Fig. 3A, B).

## Body sway during quiet stance

Four examples of stabilometric recordings performed during successive quiet-stance blocks under EC conditions, in one representative subject, are shown in the middle row of Fig. 4. The population average values of SA during the blocks of stance trials are reported in Fig. 4, separated according to visual conditions (EO, panel A; EC, panel B).

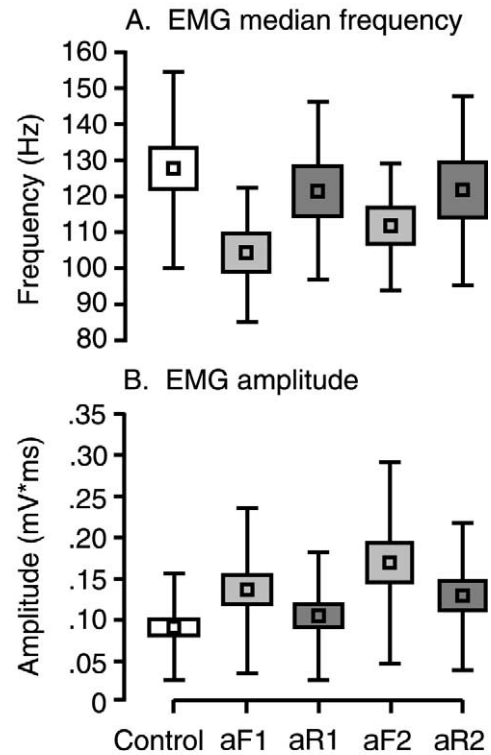


**Fig. 3.** EMG during quiet stance. Median frequency and amplitude changes of the EMG of the dorsal cervical muscles recorded during the five successive blocks of quiet-stance trials (unloaded condition). A. Median frequency slightly but significantly diminished aF1 and aF2 with respect to control and aR1 and aR2 conditions. B. Amplitude slightly but significantly increased after F1 and F2 periods, with respect to control and after-recovery conditions, with a time-course similar to that of the corresponding median frequency changes. Mean values from all subjects (±S.E.M., boxes, and ±S.D., whiskers), data from right and left muscles and EO and EC conditions collapsed.

Absolute SA values were larger with EC. Their modulations with fatigue and recovery were also more marked under EC as opposed to EO conditions. With EC, sway increased to about 120% of control values after F1 and to about 130% after F2. An incomplete trend toward control values was present after the recovery periods following F1 and F2. Sway did not fully recover to pre-fatigue value within the last recovery block. These findings were confirmed by statistical analysis. ANOVA of the body sway mean values produced a significant effect of vision (EO and EC; $F=6.613$, df 28; $P=0.016$) and sequence (blocks; $F=3.2$, df=112; $P=0.015$). The post hoc test confirmed that sway significantly increased only for the trials performed after fatigue ($P<0.01$ and $P<0.02$ for aF1 and aF2, respectively). The interaction was also significant ($F=2.55$, df=4,136; $P<0.05$), indicating that the effect of fatigue was significant only for the EC condition. In other words, when vision was allowed, the effect of the neck muscle fatiguing contractions did not significantly affect sway. The mean position of the CFP was also analysed. There were no significant effects for either visual condition or sequence of stance trials, on either antero-posterior or medio-lateral centre of pressure positions.

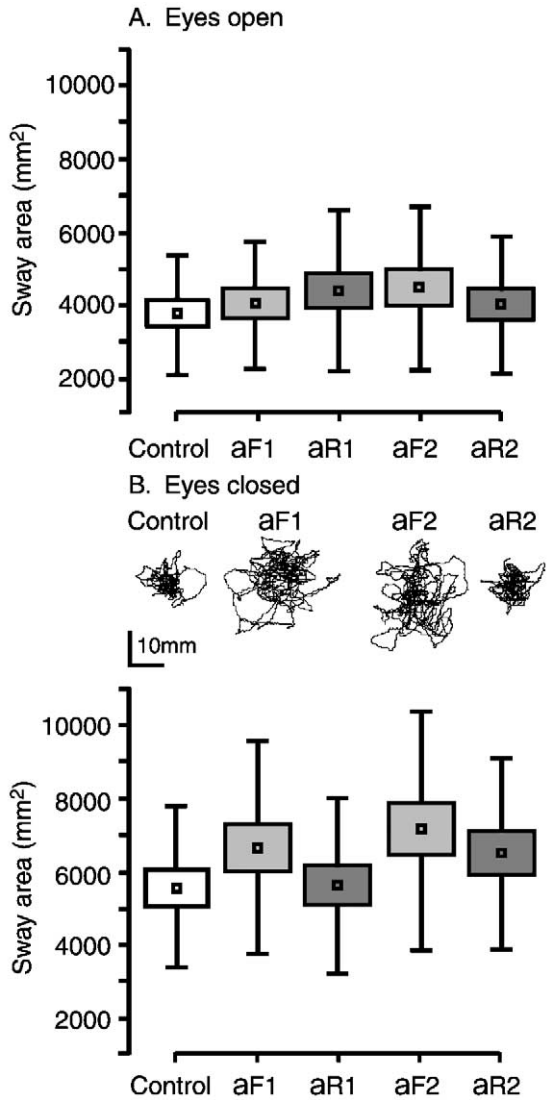


**Fig. 4.** Body sway. Stabilometric recordings during the 1-min quiet-stance acquisition periods, under control conditions and after fatiguing and recovery periods (aF1, aR1, aF2 and aR2). The plots summarise the sway values, separated according to visual conditions (A, EO; B, EC), recorded during the above sequence. The traces reported in the top row in B are the paths of CFP of a representative subject, standing with EC, under control condition and aF1, aF2 and aR2. With EC, there was a larger body sway after the fatiguing periods F1 and F2. Mean values (±S.E.M., boxes, and ±S.D., whiskers) from all subjects.

### Subjective evaluation of stability

At the end of each stance trial, subjects were asked to score how well they were able to maintain postural control (for scoring criteria, see the Experimental Procedures section). There was, as expected, a decrease in the subjective score as a function of sway worsening. However, within this general trend, there was an effect on the scoring connected with the fatiguing protocol. This is shown in Fig. 5A (left graph), which shows the plots of score versus sway under control conditions (EO and EC collapsed), after fatigue (aF1 and aF2 collapsed) and after the recovery period (aR1 and aR2 collapsed): the lines best fitting the scores given after fatigue and recovery are placed below those relative to the scores given before fatigue, for the same range of SAs. This is summarised in Fig. 5B (right histogram), which shows the mean scores given to stance trials having the same average values of log sway (control conditions 3.69±0.12 S.D.; after fatigue 3.70±0.11; after recovery 3.71±0.12; ANOVA, ns). There was a significant decrease of the mean score (two-way ANOVA, effect of fatigue, F=19.9; df 2,355; $P<0.001$), a sign of lower self-confidence in balance after fatigue. This worsening was true under both EO and EC (interaction *ns*), despite of obvious differences in sway and score between visual conditions (two-way ANOVA, effect of vision, F=120.7; df 1,355; $P<0.001$). The post hoc comparison showed that this relatively diminished self-confidence persisted after the recovery periods (post hoc test, $P<0.02$ for all conditions, except $P$=ns between aF and aR, for both EO and EC conditions).

### Non-fatiguing contractions

Six of the subjects who participated in the previously described fatiguing session took part also in a second session, several days after the former, in which the mass was reduced to about half the amount held during the fatiguing session, all other aspects being the same. This was done in order to answer the following closely related questions: a) Is a prolonged muscle contraction per se responsible for the observed effects on sway, regardless of its intensity? b) Is the particular pattern of repetition of the stance trials per se responsible for the observed effects on sway, regardless of concurrent occurrence of the phenomenon of fatigue in the neck muscles? The relevant EMG findings are summarised in Fig. 6. An index of neck muscle fatigue was obtained by dividing the median EMG frequency at the end of the contraction period (fifth minute) by the median frequency at the beginning of the same period (first minute; Fig. 6A). On average, the median frequency of the EMG signal did not significantly decrease within the 5 min of low-level neck muscle tonic contraction (for both F1 and F2), but it did so for high-level effort. Analogously, the EMG amplitude index (Fig. 6B), calculated by dividing the EMG amplitude at the end of the contraction period (fifth minute) by the amplitude at the beginning of the same period (first minute), was constant from the beginning to the end of the low-level effort. However, EMG amplitude increased when the neck was loaded with higher weights. This was true for both the first and the second 5-min contraction period.

The effects of different loads on body sway are depicted in the panel C of Fig. 6. Again, an index of increased instability was obtained by means of dividing the sway (with EC) after F1 by the sway under control conditions, and the sway after F2 by the sway after R1. An index equal to 1 meant no change occurred after neck muscle (fatiguing or non-fatiguing) contractions. When the load was reduced, there was a lesser effect of contraction on body SA, compared with high loads (two-way ANOVA; F=6.4, df 1,10; $P<0.05$). In these subjects, both loads had a significantly larger effect on sway after F2 with respect to after F1 (F=6.0, df 1,10; $P<0.05$). The subjective scores given

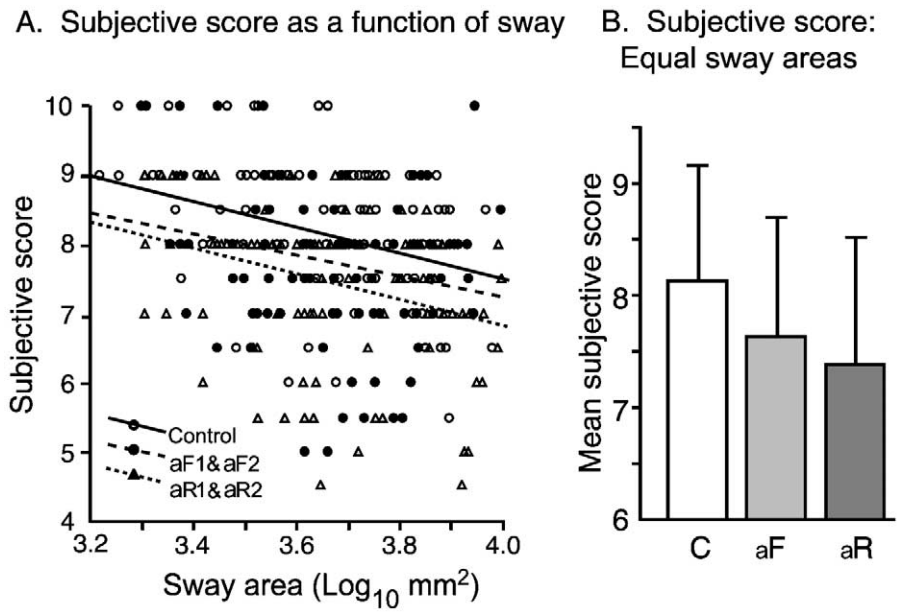


**Fig. 5.** Subjective evaluation of stance quality. A. The score given by the subjects to their stance trials (EO and EC collapsed) is plotted against the value of each corresponding trial. Under all conditions, the score diminishes with increasing sway with a similar slope. However, the lines best fitting the data show progressively lower intercepts in the order: control, after fatigue (aF1 and aF2), after recovery (aR1 and aR2). B. The three bars in the histogram report the mean scores (+S.D.) to control (C), after fatigue (aF, aF1 and F2 collapsed) and after recovery (aR, aR1 and R2 collapsed) conditions, for equal mean sway values selected across the three conditions. Subjective scoring after fatigue was significantly lower than under control conditions and did not recover after the recovery periods.

to the stance trials were not significantly altered by the low contraction level (not shown).

## DISCUSSION

Body sway during quiet stance is slightly but significantly increased after 5 min of intense contraction of dorsal neck muscles, consisting of tonic head extensor torque of about 35% of maximal voluntary torque. This protocol induced neck muscle fatigue in all subjects, as indicated by the progressive increase in EMG signal amplitude and decrease in signal frequency during contractions. These are markers of true fatigue, as indicated by many investigations on several muscle groups (Vøllestad, 1997), including neck muscle (Phillips and Petrofsky, 1983; Gogia and Sabbahi, 1994; Mannion and Dolan, 1994).

After neck muscle fatigue, the quietly standing subjects swayed more than before. This was true after both F1 and F2. Resting for 5-min periods was enough to allow recovery toward control values. Although such behaviour was detectable under both visual conditions, the fatigue effect reached statistical significance only when vision was not present. It is not unlikely that vision could have had an important stabilising effect, capable of overcoming the disturbing effects connected with the neck muscle fatigue and related abnormal sensory inflow. Vision certainly has an impact on stance control, as repeatedly shown (Paulus et al., 1984; see Redfern et al., 2001, for a recent review). Under conditions of abnormal postural muscle input, as during leaning backwards or forwards, subjects can reach more critically inclined positions with vision than when blindfolded (Schieppati et al., 1994). Under the present conditions, vision was enough to cancel the disturbing effects connected to neck muscle fatigue. Incidentally, the eyes were kept almost stationary, since subjects had a visual target in front of them. Therefore, although recording of eye movement was missing, one might suggest the retinal (visual background), rather than the extra-retinal input (proprioceptive from eye muscles; Uchida et al., 1979), to be responsible for this ‘stabilising’ influence.

It may also be noted that the effects of neck muscle fatigue upon sway might have been underestimated in the present investigation, because of the protocol characteristics, requiring several repetitions of the stance trials, which are known to induce a progressive reduction in body sway during EC stance trials (Tarantola et al., 1997). Decreases in sway may also have resulted from a gradual recovery after fatigue. However, when successive trials after F1 and F2 were compared across subjects, no significant progressive reduction in sway was recorded. Since changes in sway can be induced by moderate changes in the position of the body (Schieppati et al., 1994), any effect of neck muscle fatigue on the mean position of the CFP during stance could have indirectly produced changes in body sway. However, this was not the case in the present conditions, since no significant changes in the position of CFP were observed in the trials performed under the various visual or fatiguing conditions.

When the experimental protocol was changed to one implying less forceful neck muscle contractions, all other things being equal, EMG activity did not show signs of fatigue. Under this condition, no significant changes in body sway or subjective score of stability were observed.

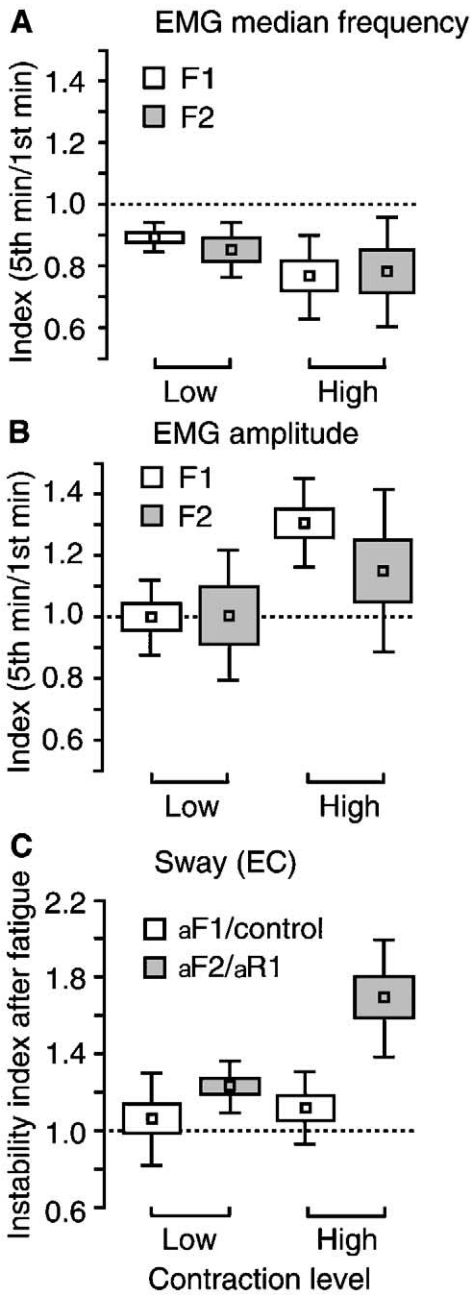


**Fig. 6.** Effect of non-fatiguing contractions. Contractions of the same duration as F1 and F2, but against smaller loads, produced minor EMG changes than the fatiguing contractions F1 and F2. The left parts and the right parts of all graphs (A–C) indicate the data obtained with non-fatiguing (Low) and fatiguing (High) loads, respectively. A and B show mean indexes of EMG median frequency and EMG amplitude, respectively, during Low and High contraction periods. The increase in body sway recorded after the Low and High contractions is reported in graph C. The sway indexes were calculated as sway after the first contraction period relative to sway under control condition, (white boxes) and sway after second contraction period relative to sway after first recovery period (grey boxes). Body sway recorded after Low contractions showed minor increases compared with sway recorded after High contractions. Mean values (±S.E.M., boxes, and ±S.D., whiskers) from the six subjects who participated in both low- and high-load sessions.

We therefore conclude that low-level contraction protocol was insufficient to cause muscle fatigue and was unable to produce major effects on quiet stance. Admittedly, general fatigue should be more pronounced during fatiguing contractions: therefore, if general fatigue had contributed, at least in part, to the effects observed, the experiments with low loads do not allow to exclude such effect. However, sizeable effects on sway have been observed after intense (exhausting uphill treadmill walking) but not moderate levels of general fatigue (Nardone et al., 1997; 1998); in the present case, instead, the fatiguing contractions did not imply a general effort of the whole body and were not equally exhausting. Another cause of increased sway could be the attentional demand connected with the act of sustaining the fatiguing contraction for 5 min, in turn contributing to general fatigue. Yet, no univocal reports exist in the literature on the possible effect of concurrent cognitive task on balance: cognitive tasks have been reported to slightly increase sway (Maki and McIlroy, 1996), to have no effect (Yardley et al., 1999) or even decrease sway (Andersson et al., 2002) in normal young subjects.

The occurrence of central fatigue cannot be excluded, either. Indeed, the 5 min fatiguing contraction was close to the sustainable limit, a condition likely to produce fatigue in the central pathway for the neck muscle activation (see Gandevia, 2001). Central fatigue might therefore interfere with the control of balance during quiet stance, but our protocol did not allow us to draw conclusions in this sense.

Therefore, we would argue that localised neck muscle fatigue may affect central mechanisms of postural control by producing an input to the CNS, associated with regional changes accompanying fatigue. Such modifications to sensory input may be the result of an increased inflow from the free nerve endings as a consequence of ionic or metabolic changes (elevated interstitial potassium concentration or insufficient oxygen availability due to reduced blood flow; Garland, 1991; Pettorossi et al., 1999). Interestingly, cervical pain-related input is able to significantly alter postural control (Karlberg et al., 1995), and neck nociceptive input has been recently shown to induce changes in the ability to correctly perceive verticality (Grod and Diakow, 2002). The present results do not permit us, however, to propose any direct mechanism responsible for abnormal small-fibre sensory input. On the other hand, it is known that fatigue can affect the discharge of many sensory receptors, including muscle spindles (Johansson and Sojka, 1991; Hill, 2001; Taylor et al., 2000; Weerakkody et al., 2001). In turn, spindle discharge could directly be affected by fatigue (Zhang and Rymer, 2001), or modulated by fatigue-induced alterations in $\gamma$-motoneurone discharge (Ljubisavljevic and Anastasijevic, 1996; Pedersen et al., 1998). Such abnormal muscle spindle inflow could possibly perturb and deteriorate postural control.

After the second fatiguing contraction (F2), body sway was increased more than after the first one, though not significantly so in the whole population. It is possible that the concentration of metabolites presumed to be responsible for abnormal receptor stimulation had reached a threshold, or that some time period was necessary in order

for a receptor sensitisation effect to occur (Hayward et al., 1991). As a non-alternative explanation, a decreased motivation of some subjects could have affected the quality of their stance after the F2, as a correlate of general fatigue.

The presence of abnormal sensory input connected with the fatiguing contractions could be indicated by the different degree of perception of self-stability reported by subjects after the fatiguing periods. When, at the end of each stance trial, subjects self-rated their performance, their increased sway was accompanied by a worse score, as expected by the previously described robust match between sway and score (Schieppati et al., 1999). The score was, however, worse than that expected for the same sway before fatigue. Within an ample range of sways, the significant decrease of half a point indicated an average worse scoring for equal sway, after rather than before fatigue. This may be another indication of true fatigue, since prolonged but non-fatiguing contractions do not produce abnormal kinaesthetic sensations (Stelmach and Worringham, 1985). This is in keeping with the hypothesis that neck muscle fatigue may be directly responsible for abnormal inputs disturbing the subjective postural reference. After recovery, scoring seemed not to recuperate toward control score rates. A simple supposition is that a dissociation occurred in the central effects of fatigue-induced sensory inflow: the capacity to reflexively control sway recovered earlier than the perceptual disturbance. The latter would be connected with a more persistent altered reconfiguration of the internal reference (Karnath et al., 2002), or to a more 'fragile' nervous mechanism responsible for sway perception, liable to be easily disturbed by minimal alterations in neck sensory inflow persisting after recovery, or both. Interestingly, correct perception of sway was shown to be unaffected by either peripheral neuropathy or Parkinsonism (Schieppati et al., 1999). Then, the afferent discharge from the neck might play a dominant role in calibrating all other sensory inputs; an anomalous inflow and the consequent incongruent central integration of it might be one of the factors contributing to dizziness.

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## REFERENCES

Alund M, Ledin T, Odkvist L, Larsson SE (1993) Dynamic posturography among patients with common neck disorders: a study of 15 cases with suspected cervical vertigo. J Vest Res 3:383–389.

Andersson G, Hagman J, Talianzadeh R, Svedberg A, Larsen HC (2002) Effect of cognitive load on postural control. Brain Res Bull 58:135–139.

Beretta MV, Bove M, Burroni E, Diverio M, Facioli M, Nardone A, Schieppati M (2002) Neck muscle fatigue affects postural control. Gait Posture 16 (Suppl 1):S227.

Biguer B, Donaldson IM, Hein A, Jeannerod M (1988) Neck muscle vibration modifies the representation of visual motion and direction in man. Brain 111:1405–1424.

Bove M, Courtine G, Schieppati M (2002) Neck muscle vibration and spatial orientation during stepping in place in humans. J Neurophysiol 88:2232–2241.

Boyle R, Pompeiano O (1981) Relation between cell size and response characteristics of vestibulospinal neurons to labyrinth and neck inputs. J Neurosci 1:1052–1066.

Brandt T, Bronstein AM (2001) Cervical vertigo. J Neurol Neurosurg Psychiatr 71:8–12.

Carpenter JE, Blasier RB, Pellizzon GG (1998) The effects of muscle fatigue on shoulder joint position sense. Am J Sports Med 26:262–265.

Endo K, Ichimaru K, Shimura H, Imakiire A (2000) Cervical vertigo after hair shampoo treatment at a hairdressing salon: a case report. Spine 25:632–634.

Fransson PA, Karlberg M, Sterner T, Magnusson M (2000) Direction of galvanically-induced vestibulo-postural responses during active and passive neck torsion. Acta Otolaryngol 120:500–503.

Gandevia SC (2001) Spinal and supraspinal factors in human muscle fatigue. Physiol Rev 81:1725–1789.

Garland SJ (1991) Role of small diameter afferents in reflex inhibition during human muscle fatigue. J Physiol 435:547–558.

Gdowski GT, McCrea RA (2000) Neck proprioceptive inputs to primate vestibular nucleus neurons. Exp Brain Res 135:511–526.

Gogia PP, Sabbahi MA (1994) Electromyographic analysis of neck muscle fatigue in patients with osteoarthritis of the cervical spine. Spine 19:502–506.

Goodwin GM, McCloskey DI, Matthews PB (1972) The contribution of muscle afferents to kinaesthesia shown by vibration induced illusions of movement and by the effects of paralysing joint afferents. Brain 1972:705–748.

Grod JP, Diakow PR (2002) Effect of neck pain on verticality perception: a cohort study. Arch Physiol Med Rehab 83:412–415.

Gurfinkel VS, Popov KE, Smetanin BN, Shlykov VY (1989) Changes in the direction of vestibulo-motor response in the course of adaptation to protracted static head turning in man. Neurophysiology 21:159–164.

Hayward L, Wesselmann U, Rymer WZ (1991) Effects of muscle fatigue on mechanically sensitive afferents of slow conduction velocity in the cat triceps surae. J Neurophysiol 65:360–370.

Hill JM (2001) Increase in the discharge of muscle spindles during diaphragm fatigue. Brain Res 918:166–170.

Johansson H, Sojka P (1991) Pathophysiological mechanisms involved in the genesis and spread of muscular tension in occupational muscle pain and in chronic musculoskeletal pain syndromes: a hypothesis. Med Hypoth 35:196–203.

Karlberg M, Johansson R, Magnusson M, Fransson PA (1996) Dizziness of suspected cervical origin distinguished by posturographic assessment of human postural dynamics. J Vestib Res 6:37–47.

Karlberg M, Persson L, Magnusson M (1995) Reduced postural control in patients with chronic cervicobrachial pain syndrome. Gait Posture 3:241–249.

Karnath HO, Sievering D, Fetter M (1994) The interactive contribution of neck muscle proprioception and vestibular stimulation to subjective "straight ahead" orientation in man. Exp Brain Res 101:140–146.

Karnath HO, Reich E, Rorden C, Fetter M, Driver J (2002) The perception of body orientation after neck-proprioceptive stimulation: effects of time and of visual cueing. Exp Brain Res 143:350–358.

Kavounoudias A, Gilhodes JC, Roll R, Roll JP (1999) From balance regulation to body orientation: two goals for muscle proprioceptive information processing? Exp Brain Res 124:80–88.

Koskimies K, Sutinen P, Aalto H, Starck J, Toppila E, Hirvonen T, Kaksonen R, Ishizaki H, Alaranta H, Pyykko I (1997) Postural stability, neck proprioception and tension neck. Acta Otolaryngol Suppl 529:95–97.

Ljubisavljevic M, Anastasijevic R (1996) Fusimotor system in muscle fatigue. J Periph Nerv Syst 1:83–96.

Maki BE, McIlroy WE (1996) Influence of arousal and attention on the control of postural sway. J Vestib Res 6:53–59.

Mannion AF, Dolan P (1994) Electromyographic median frequency changes during isometric contraction of the back extensors to fatigue. Spine 19:1223–1229.

Mazzini L, Schieppati M (1994) Short-latency neck muscle responses to vertical body tilt in normal subjects and in patients with spasmodic torticollis. Electroencephalogr Clin Neurophysiol 93:265–275.

Mergner T, Siebold C, Schweigart G, Becker W (1991) Human perception of horizontal trunk and head rotation in space during vestibular and neck stimulation. Exp Brain Res 85:389–404.

Mergner T, Rottler G, Kimmig H, Becker W (1992) Role of vestibular and neck inputs for the perception of object motion in space. Exp Brain Res 89:655–668.

Mergner T, Huber W, Becker W (1997) Vestibular-neck interaction and transformation of sensory coordinates. J Vestib Res 7:347–367.

Merletti R, Lo Conte LR (1997) Surface EMG signal processing during isometric contractions. J Electromyogr Kinesiol 7:241–250.

Nardone A, Tarantola J, Giordano A, Schieppati M (1997) Fatigue effects on body balance. Electroenceph Clin Neurophysiol 105: 309–320.

Nardone A, Tarantola J, Galante M, Schieppati M (1998) Time course of stabilometric changes after a strenuous treadmill exercise. Arch Phys Med Rehab 79:920–924.

Paulus WM, Straube A, Brandt T (1984) Visual stabilization of posture: physiological stimulus characteristics and clinical aspects. Brain 107:1143–1163.

Pedersen J, Ljubisavljevic M, Bergenheim M, Johansson H (1998) Alterations in information transmission in ensembles of primary muscle spindle afferents after muscle fatigue in heteronymous muscle. Neuroscience 84:953–959.

Pettorossi VE, Della Torre G, Bortolami R, Brunetti O (1999) The role of capsaicin-sensitive muscle afferents in fatigue-induced modulation of the monosynaptic reflex in the rat. J Physiol 515:599–607.

Phillips CA, Petrofsky JS (1983) Quantitative electromyography: response of the neck muscles to conventional helmet loading. Aviat Space Environ Med 54:452–457.

Redfern MS, Yardley L, Bronstein AM (2001) Visual influences on balance. J Anxiety Disord 15:81–94.

Roll R, Velay JL, Roll JP (1991) Eye and neck proprioceptive messages contribute to the spatial coding of retinal input in visually oriented activities. Exp Brain Res 85:423–431.

Schieppati M, Grasso M, Siliotto R, Nardone A (1993) Effect of age, chronic diseases and Parkinsonism on postural control. In: Sensorimotor impairment in the elderly (Stelmach GE, Hömberg V, eds), pp 355–373. The Netherlands: Kluwer.

Schieppati M, Hugon M, Grasso M, Nardone A, Galante M (1994) The limits of equilibrium in young and elderly normal subjects and in parkinsonians. Electroenceph Clin Neurophysiol 93:286–298.

Schieppati M, Tacchini E, Nardone A, Tarantola J, Corna S (1999) Subjective perception of body sway. J Neurol Neurosurg Psychiatr 66:313–322.

Sharpe MH, Miles TS (1993) Position sense at the elbow after fatiguing contractions. Exp Brain Res 94:179–182.

Stelmach GE, Worringham CJ (1985) Sensorimotor deficits related to postural stability: implications for falling in the elderly. Clin Geriatr Med 1:679–694.

Taimela S, Kankaanpaa M, Luoto S (1999) The effect of lumbar fatigue on the ability to sense a change in lumbar position: a controlled study. Spine 24:1322–1327.

Tarantola J, Nardone A, Tacchini E, Schieppati M (1997) Human stance stability improves with the repetition of the task: effect of foot position and visual condition. Neurosci Lett 228:75–78.

Taylor JL, McCloskey DI (1991) Illusions of head and visual target displacement induced by vibration of neck muscles. Brain 114:755–759.

Taylor JL, Butler JE, Gandevia SC (2000) Changes in muscle afferents, motoneurons and motor drive during muscle fatigue. Eur J Appl Physiol 83:106–115.

Uchida T, Hashimoto M, Suzuki N, Takegami T, Iwase Y (1979) Effects of periodic saccades on the body sway in human subjects. Neurosci Lett 13:253–258.

Vøllestad NK (1997) Measurement of human muscle fatigue. J Neurosci Methods 74:219–227.

Weerakkody N, Whitehead NP, Canny BJ, Gregory JE, Proske U (2001) Large-fiber mechanoreceptors contribute to muscle soreness after eccentric exercise. J Pain 2:209–219.

Yardley L, Gardner M, Leadbetter A, Lavie N (1999) Effect of articulatory and mental tasks on postural control. Neuroreport 10:215–219.

Yoshitake Y, Ue H, Miyazaki M, Moritani T (2001) Assessment of lower-back muscle fatigue using electromyography, mechanomyography, and near-infrared spectroscopy. J Appl Physiol 84:174–179.

Zhang LQ, Rymer WZ (2001) Reflex and intrinsic changes induced by fatigue of human elbow extensor muscles. J Neurophysiol 86:1086–1094.

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# CORRIGENDUM

## CORRIGENDUM TO "NECK MUSCLE FATIGUE AFFECTS POSTURAL CONTROL IN MAN" [Neuroscience 121 (2) 277–285]

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The authors regret the following error.

On page 279, left column, lines 2–5, the sentence should read:

"For each subject, the mass was chosen so that it represented about one third of the maximal voluntary contraction (MVC, 23.5±5.5 kg), i.e. 8.2 kg±2.7 S.D., equal to 34.9±4.8% of MVC)."

This error did not change the message of the paper, since all the forces produced by the subjects during their contractions were expressed and analyzed in the paper as percentage of the maximal voluntary contraction, rather than in absolute terms. Moreover, the fact that the contractions did produce muscle fatigue was not based on the absolute load values sustained, but rather on the analysis of the EMG amplitude and median frequency.

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